

Hi-C for Chromosome Conformation Capture Technologies

April 18 2026

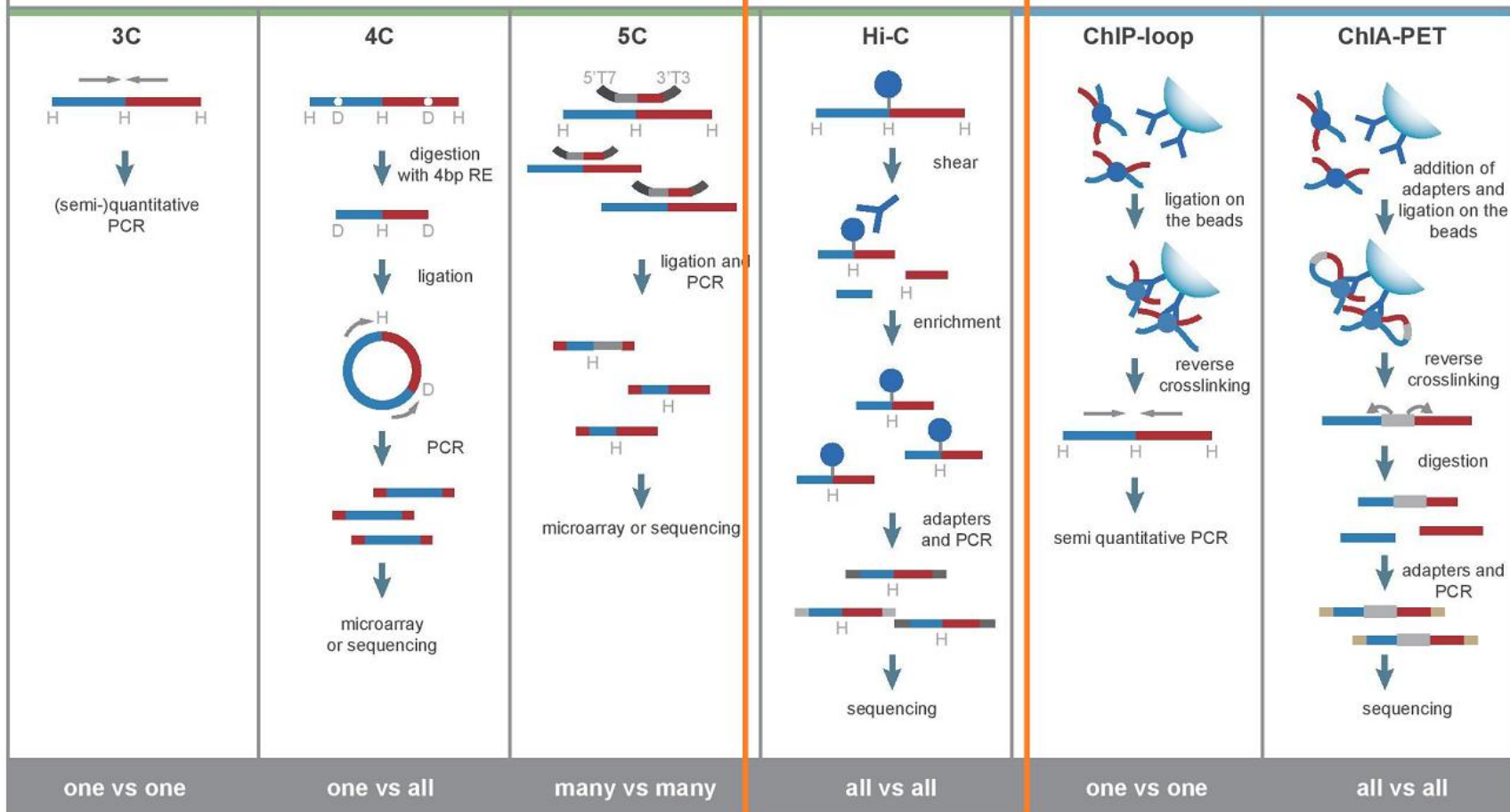
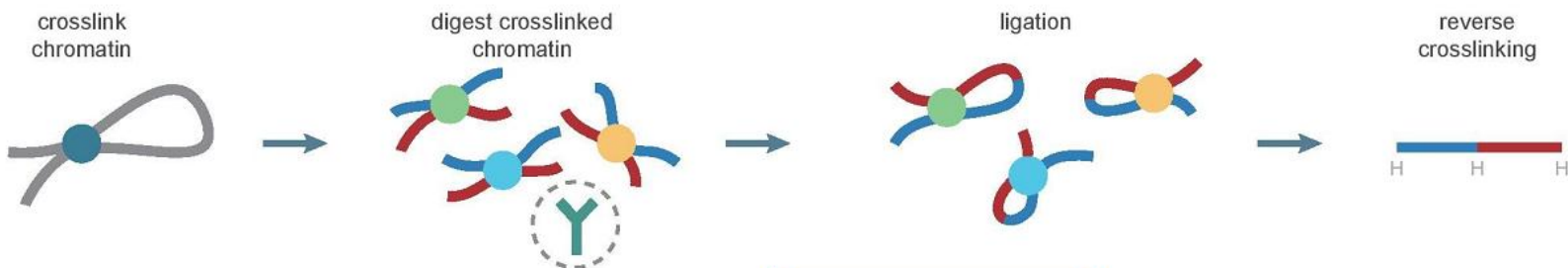
Phuc-Loi Luu, PhD

Content

- 1. From 3C, 4C, 5C to Hi-C**
- 2. Reading Hi-C output**



Chromosome Conformation Technologies



Hi-C



Comprehensive Mapping of Long-Range Interactions Reveals Folding Principles of the Human Genome

Erez Lieberman-Aiden *et al.*

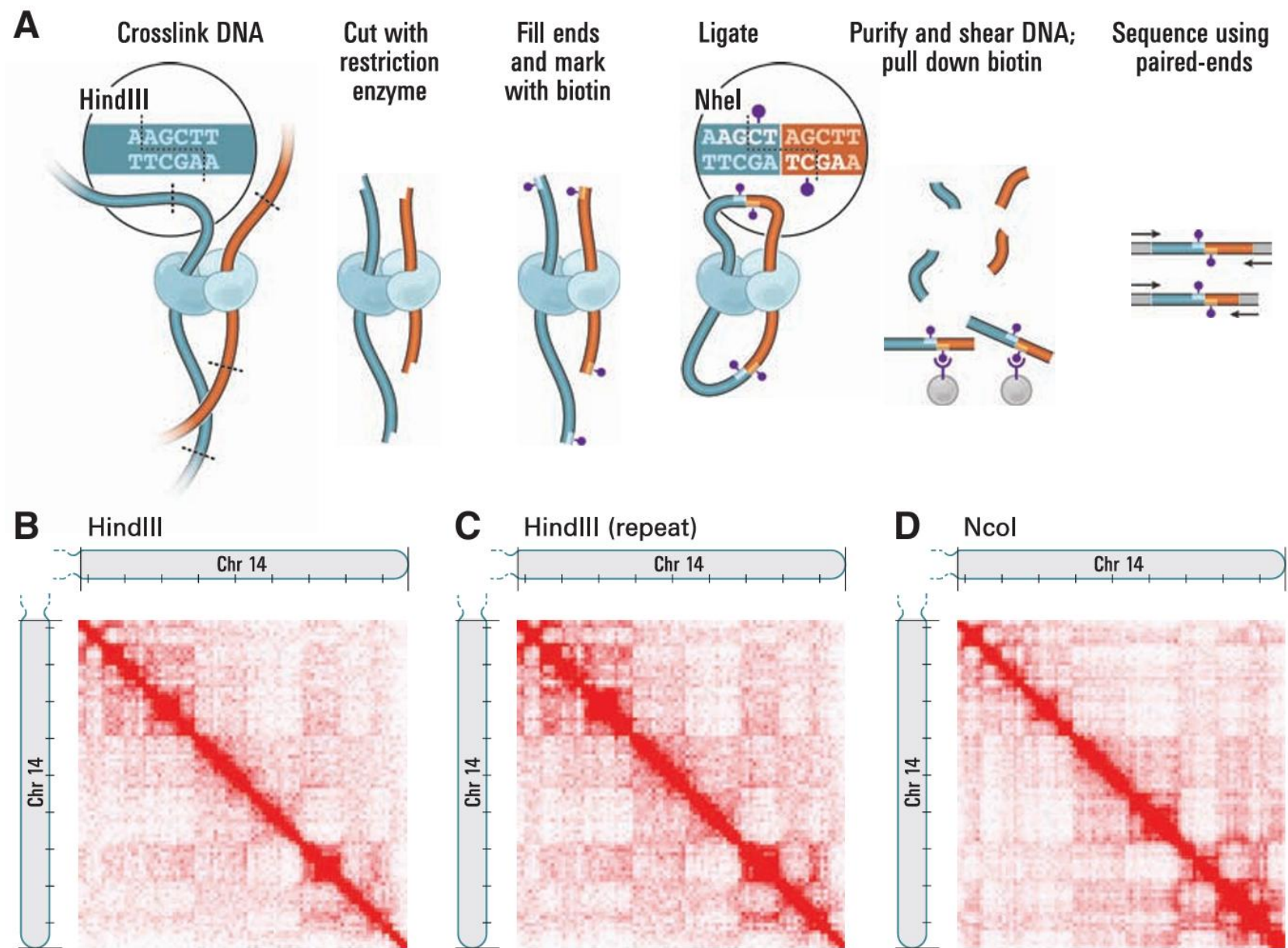
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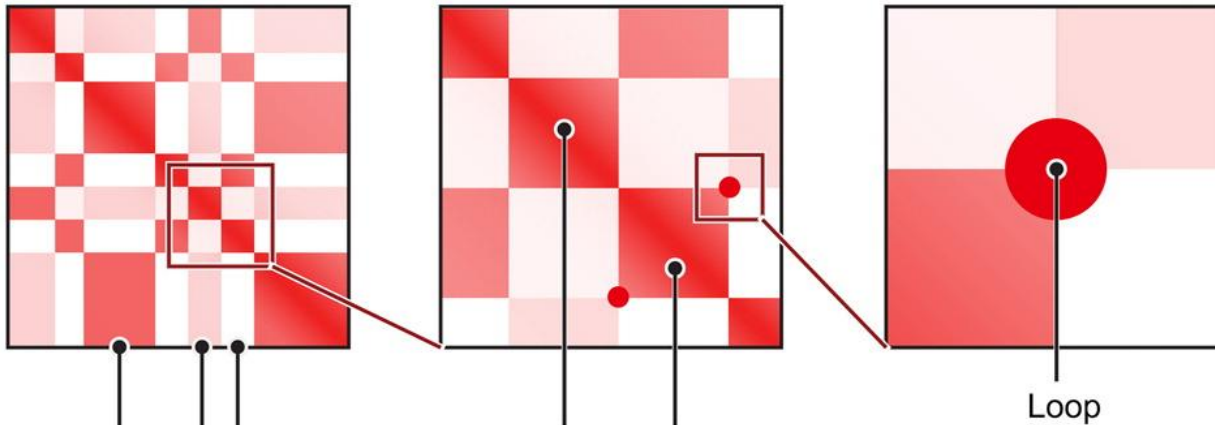
We describe Hi-C, a method that probes the three-dimensional architecture of whole genomes by coupling proximity-based ligation with massively parallel sequencing. We constructed spatial proximity maps of the human genome with Hi-C at a resolution of 1 megabase. These maps confirm the presence of chromosome territories and the spatial proximity of small, gene-rich chromosomes. We identified an additional level of genome organization that is characterized by the spatial segregation of open and closed chromatin to form two genome-wide compartments. At the megabase scale, the chromatin conformation is consistent with a fractal globule, a knot-free, polymer conformation that enables maximally dense packing while preserving the ability to easily fold and unfold any genomic locus. The fractal globule is distinct from the more commonly used globular equilibrium model. Our results demonstrate the power of Hi-C to map the dynamic conformations of whole genomes.

Fig. 1. Overview of Hi-C. **(A)** Cells are cross-linked with formaldehyde, resulting in covalent links between spatially adjacent chromatin segments (DNA fragments shown in dark blue, red; proteins, which can mediate such interactions, are shown in light blue and cyan). Chromatin is digested with a restriction enzyme (here, HindIII; restriction site marked by dashed line; see inset), and the resulting sticky ends are filled in with nucleotides, one of which is biotinylated (purple dot). Ligation is performed under extremely dilute conditions to create chimeric molecules; the HindIII site is lost and an NheI site is created (inset). DNA is purified and sheared. Biotinylated junctions are isolated with streptavidin beads and identified by paired-end sequencing. **(B)** Hi-C produces a genome-wide contact matrix. The submatrix shown here corresponds to intrachromosomal interactions on chromosome 14. (Chromosome 14 is acrocentric; the short arm is not shown.) Each pixel represents all interactions between a 1-Mb locus and another 1-Mb locus; intensity corresponds to the total number of reads (0 to 50). Tick marks appear every 10 Mb. **(C and D)** We compared the original experiment with results from a biological repeat using the same restriction enzyme [(C), range from 0 to 50 reads] and with results using a different restriction enzyme [(D), NcoI, range from 0 to 100 reads].

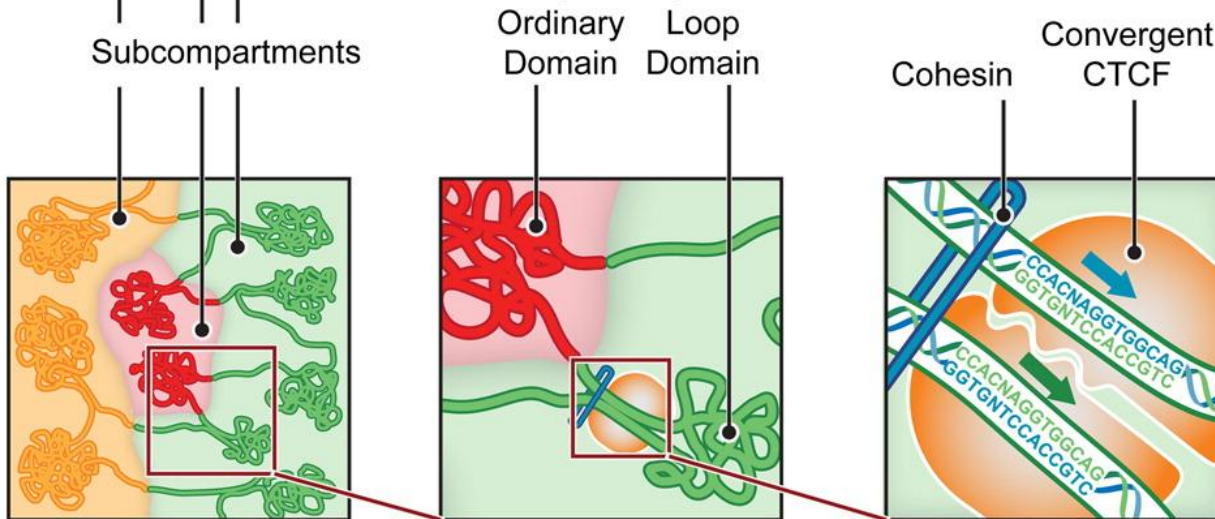


Hi-C Matrices and Models

Matrices



Models



<https://www.cell.com/fulltext/S0092-8674%2814%2901497-4>

A

3D Genome Architecture

Hi-C

Genome

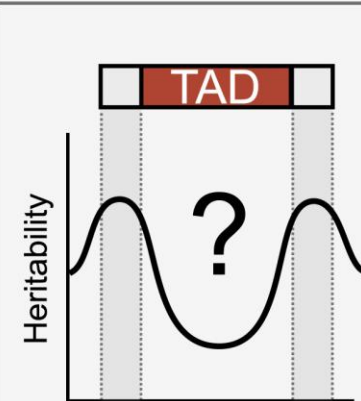
Boundary

TAD

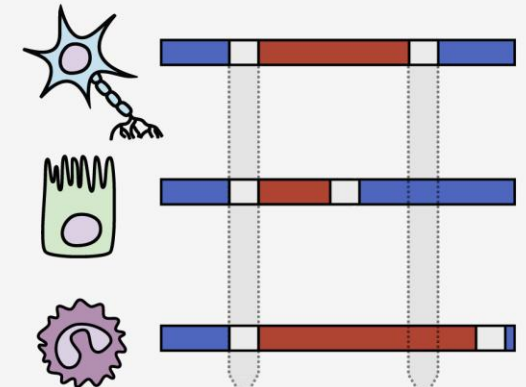
TAD

B

Comparisons

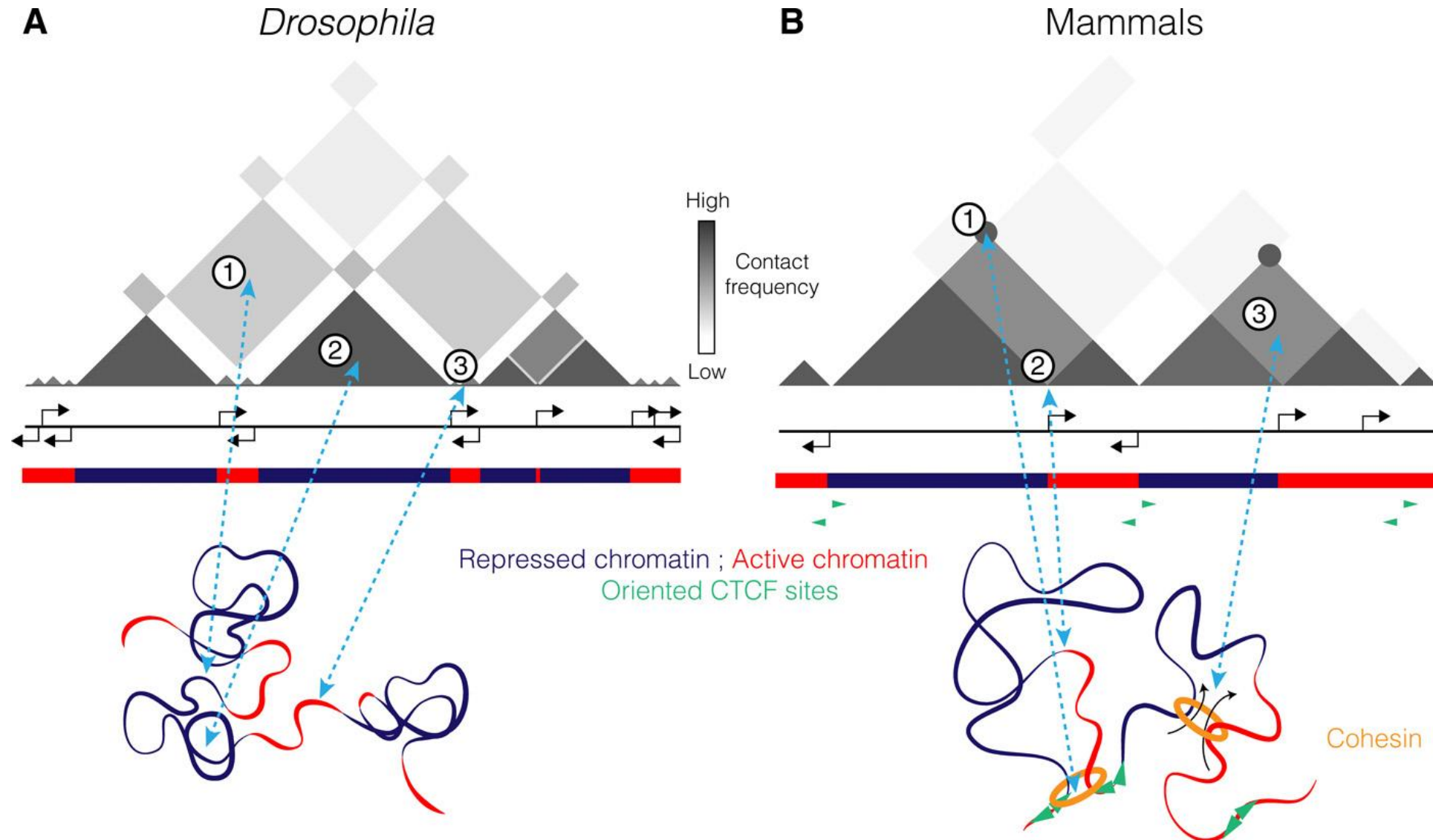


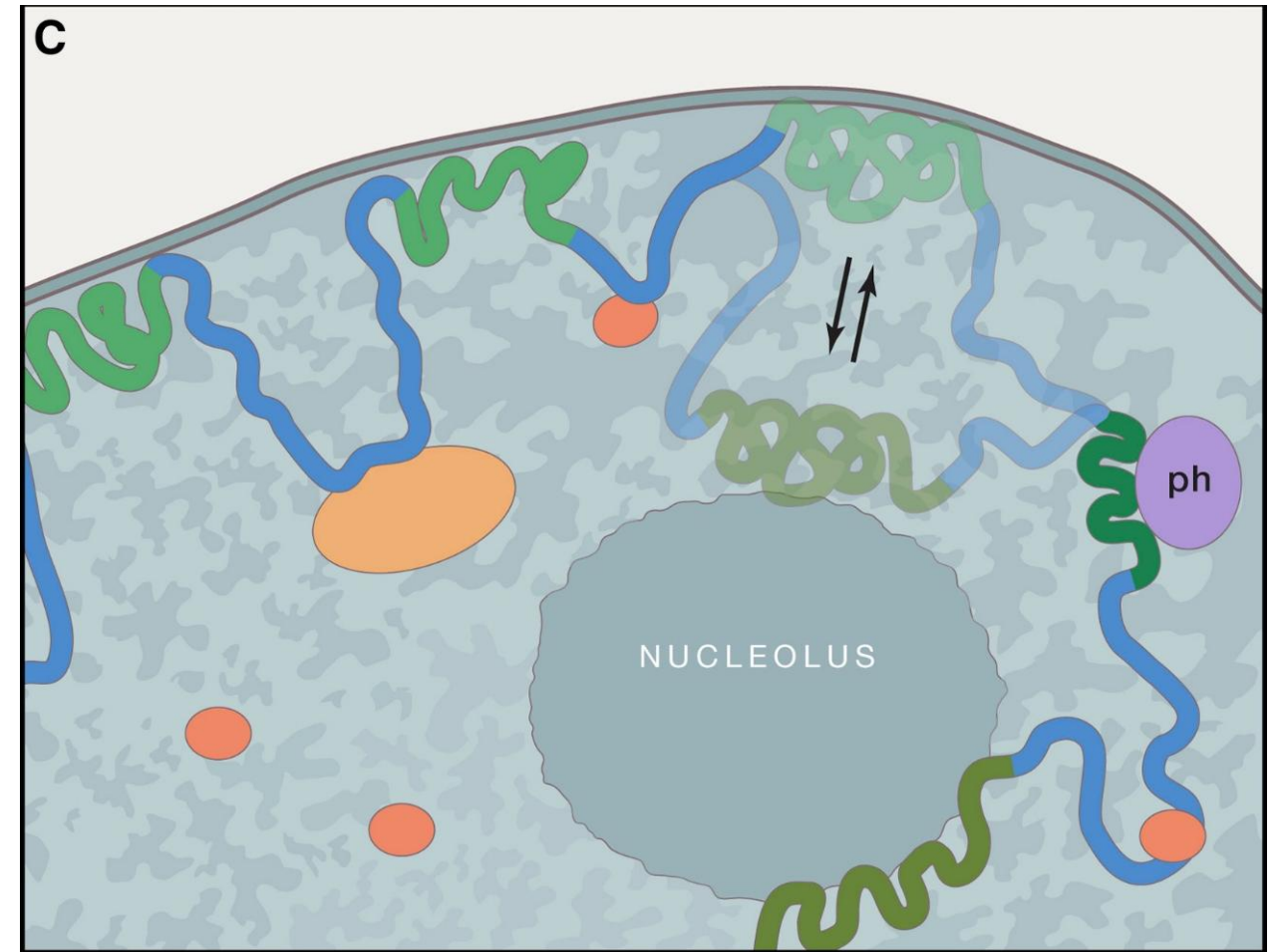
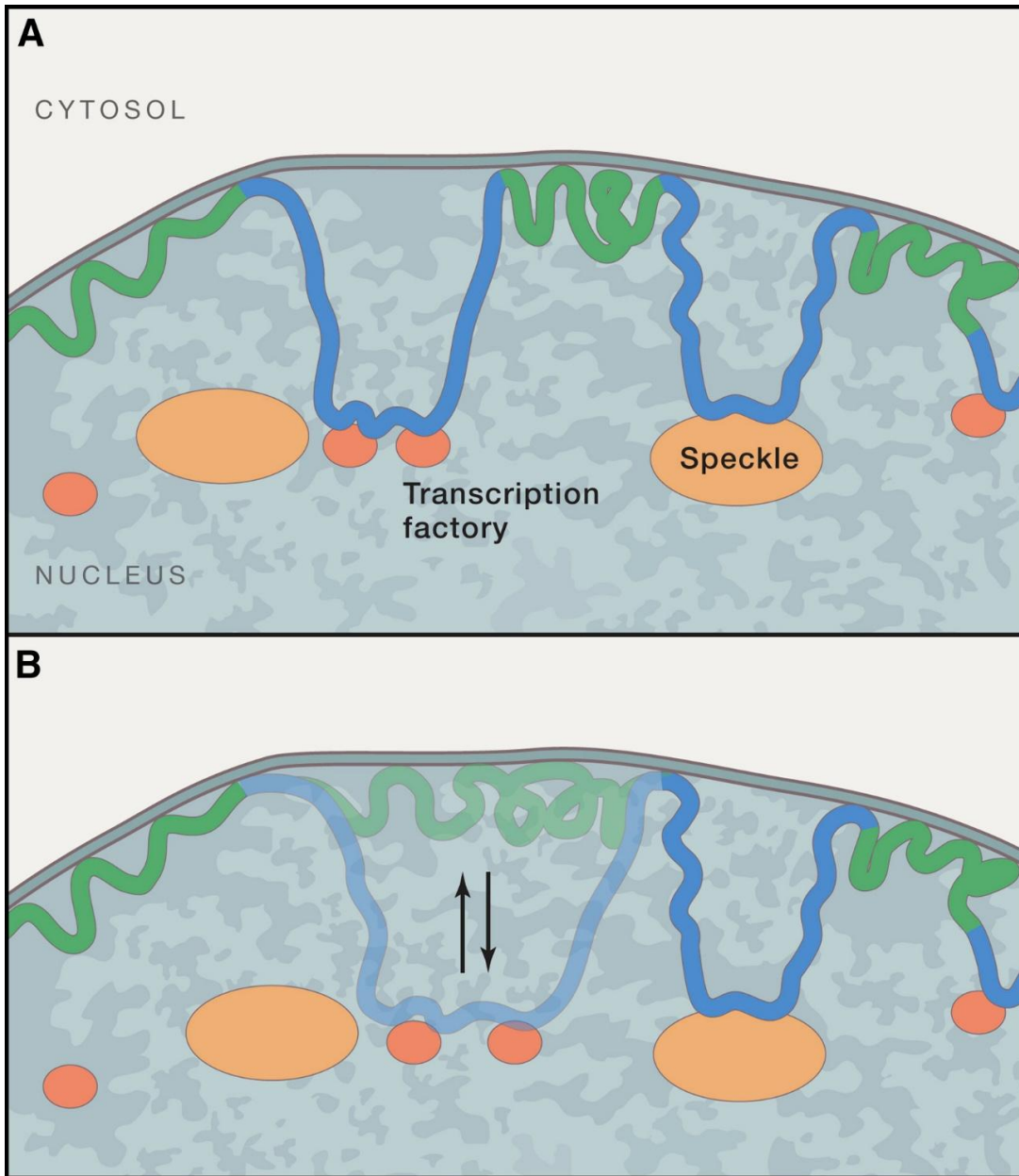
TAD vs. boundary



<https://www.sciencedirect.com/science/article/pii/S000292972100001X>

Read the Hi-C data (0)



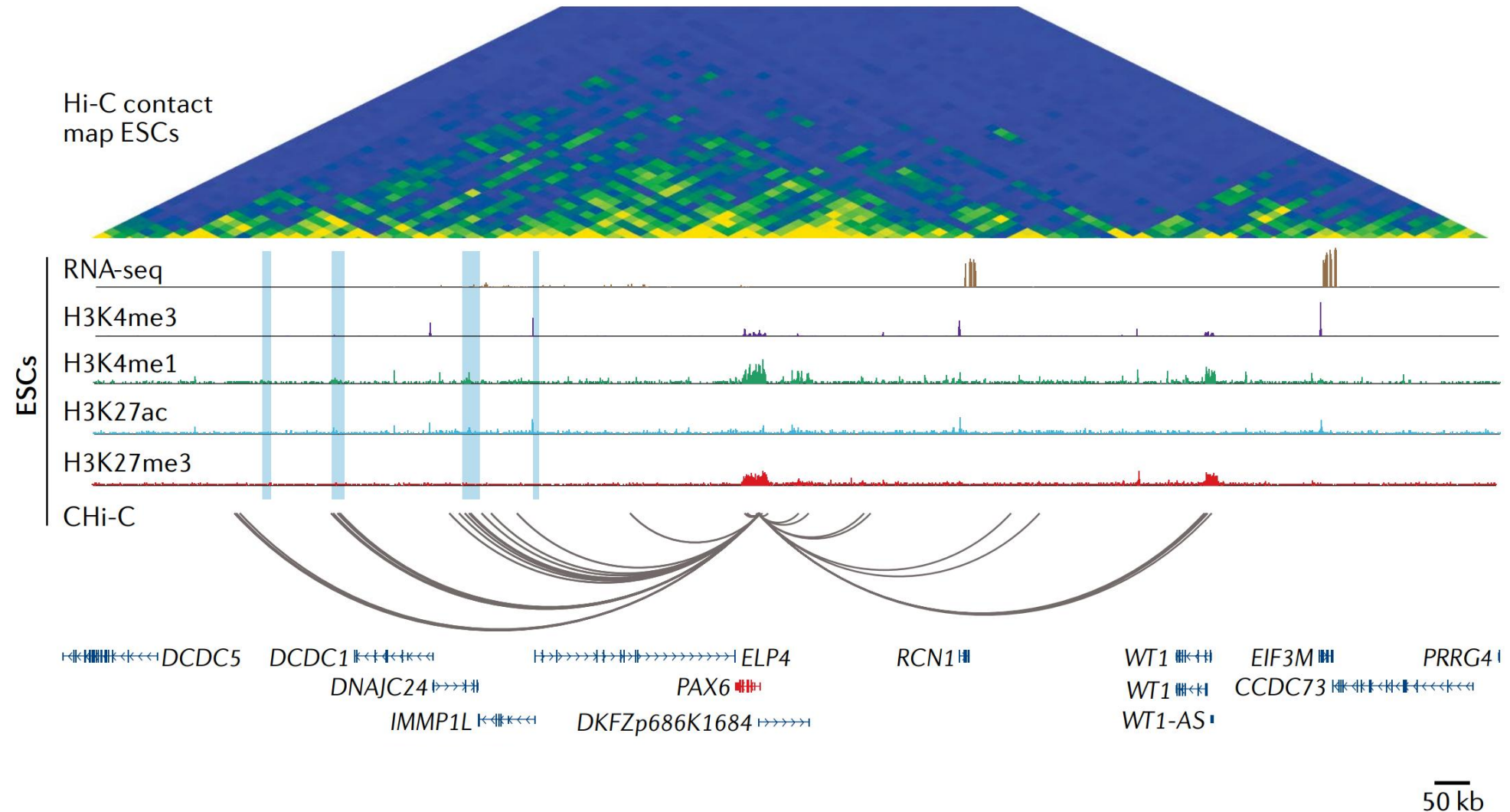


(A) Besides anchoring of LADs (green) to the NL, other regions (blue) may be tethered to nuclear structures that are permissive for transcription (orange), such as transcription factories (tf) or splicing factor speckles (speckles).

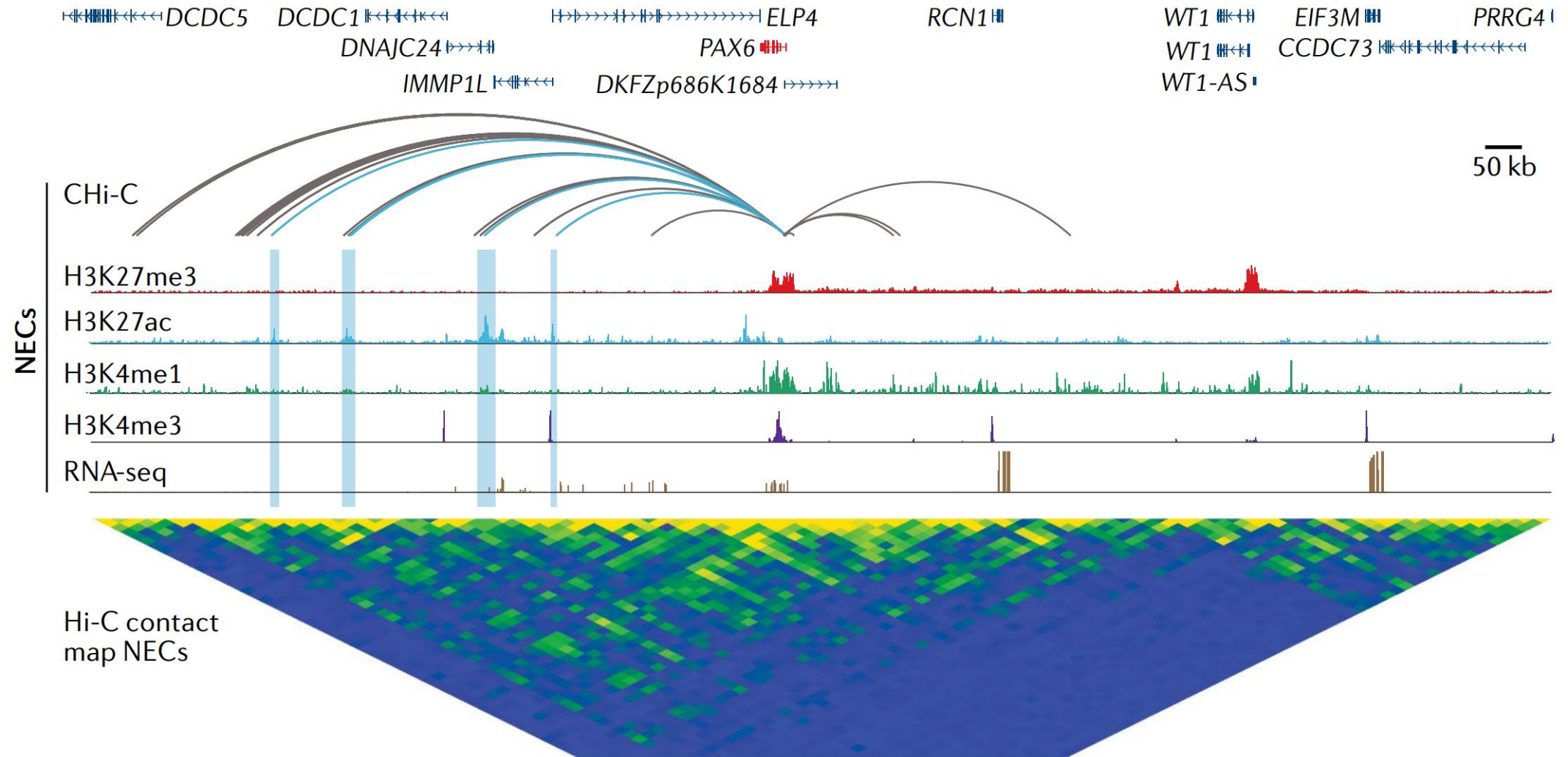
(B) Some LADs (semi-transparent green) contact the NL erratically (i.e., in a subset of cells) and may become transcriptionally active when associated with a permissive compartment (semi-transparent blue).

(C) Some LADs are apparently stochastically distributed between the NL, nucleoli, and pericentromeric heterochromatin (ph), which are all repressive environments.

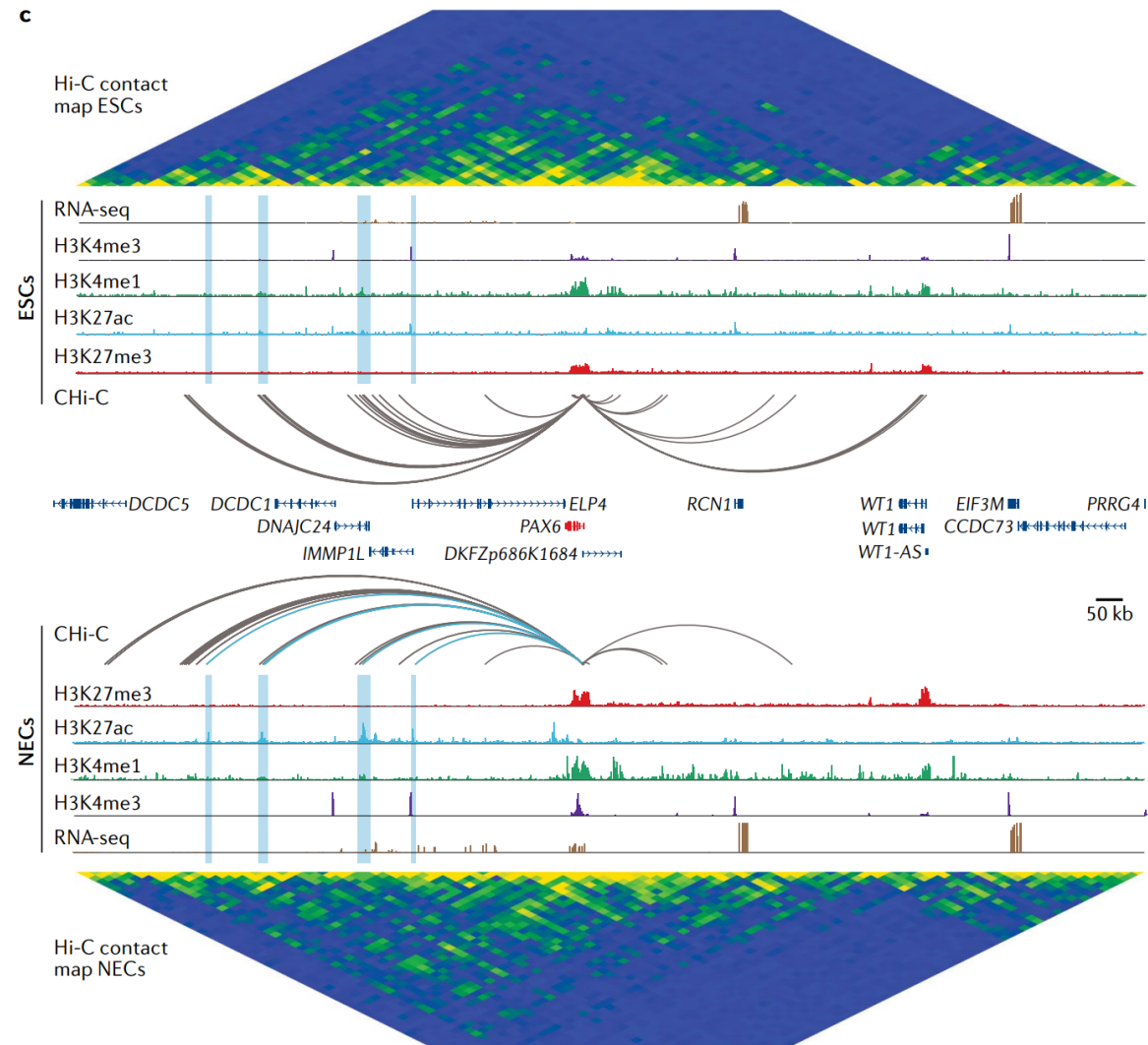
Read the Hi-C data (1)



Read the Hi-C data (2)



Read the Hi-C data (3)



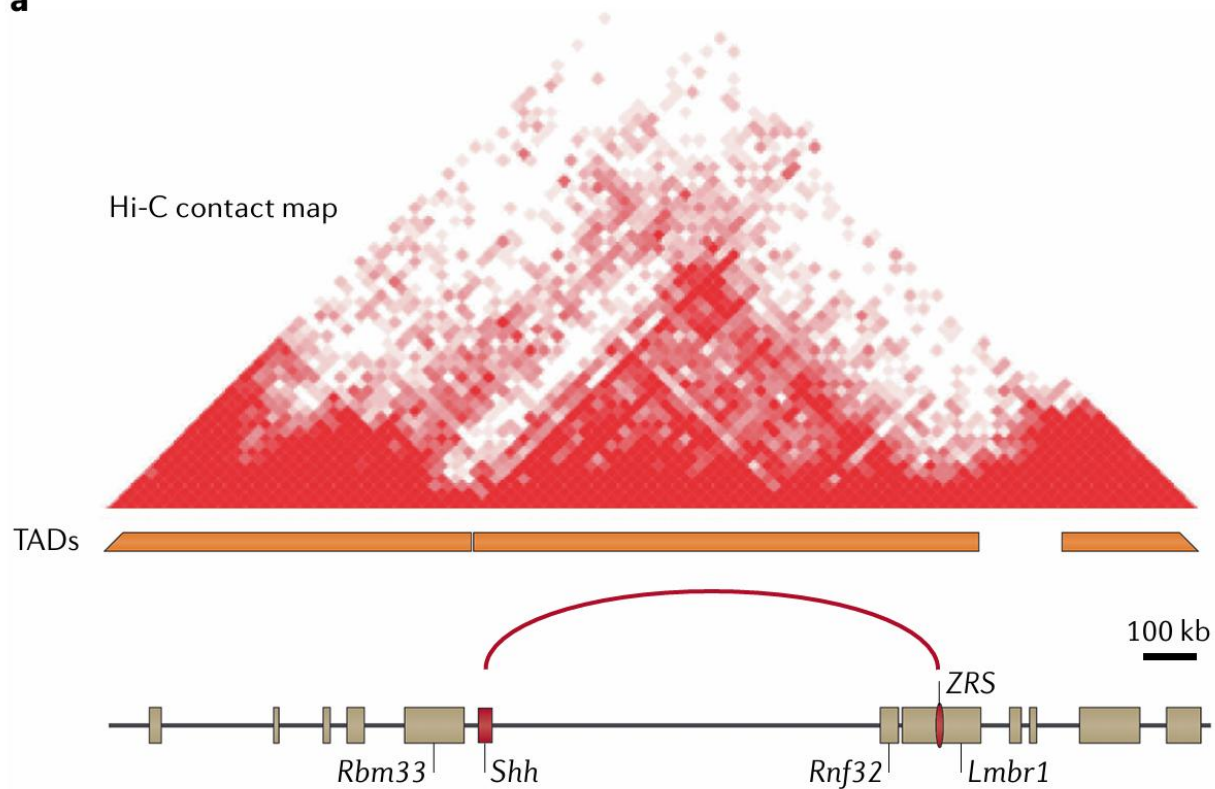
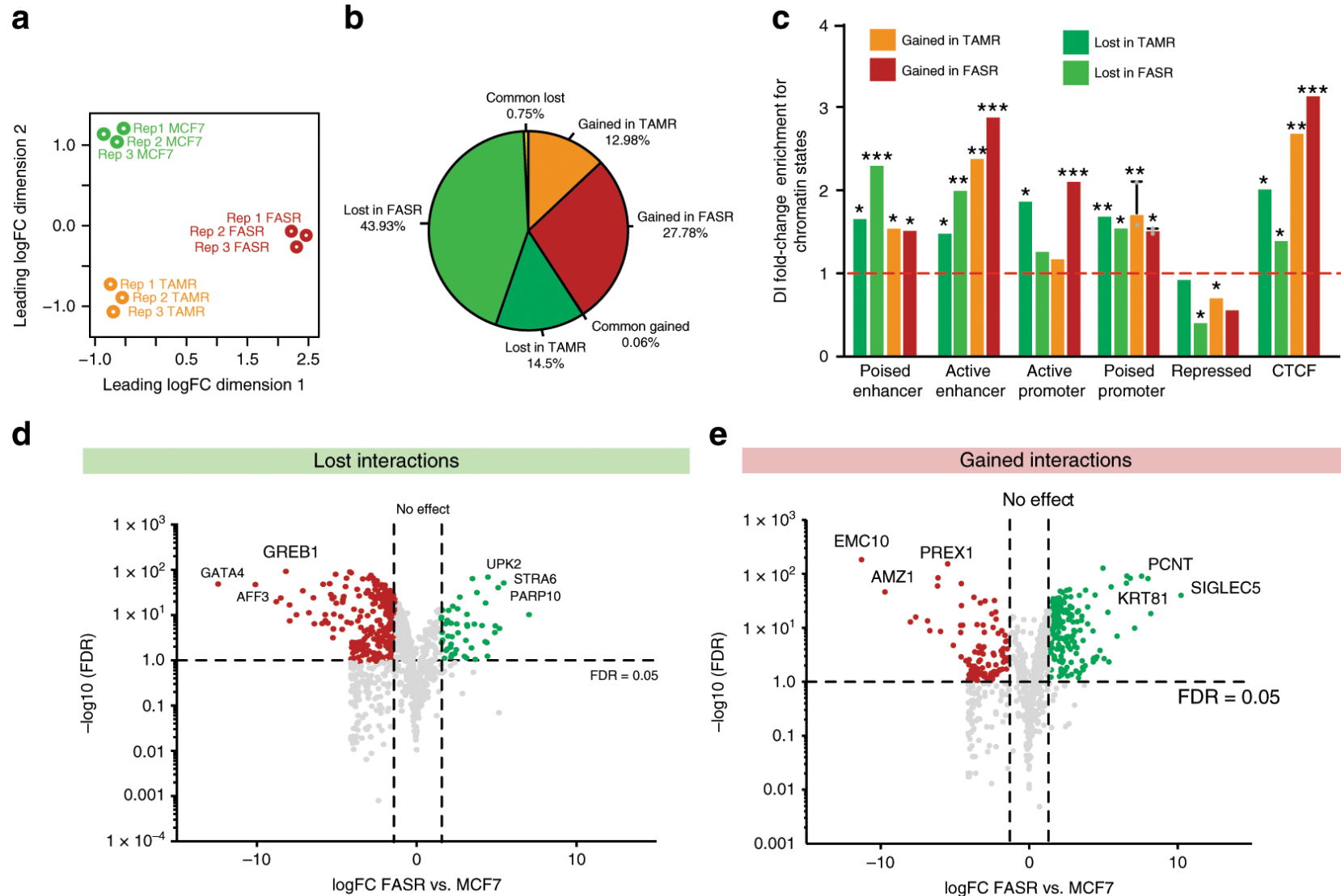
a**b**

Fig. 2 | The *Shh* limb bud enhancer: a paradigm for long-range enhancer control of gene expression.

a | Genomic organization of the mouse sonic hedgehog (*Shh*) locus. In developing limb buds, the ‘zone of polarizing activity regulatory sequence’ (ZRS) enhancer controls the expression of *Shh* located ~850 kb away.

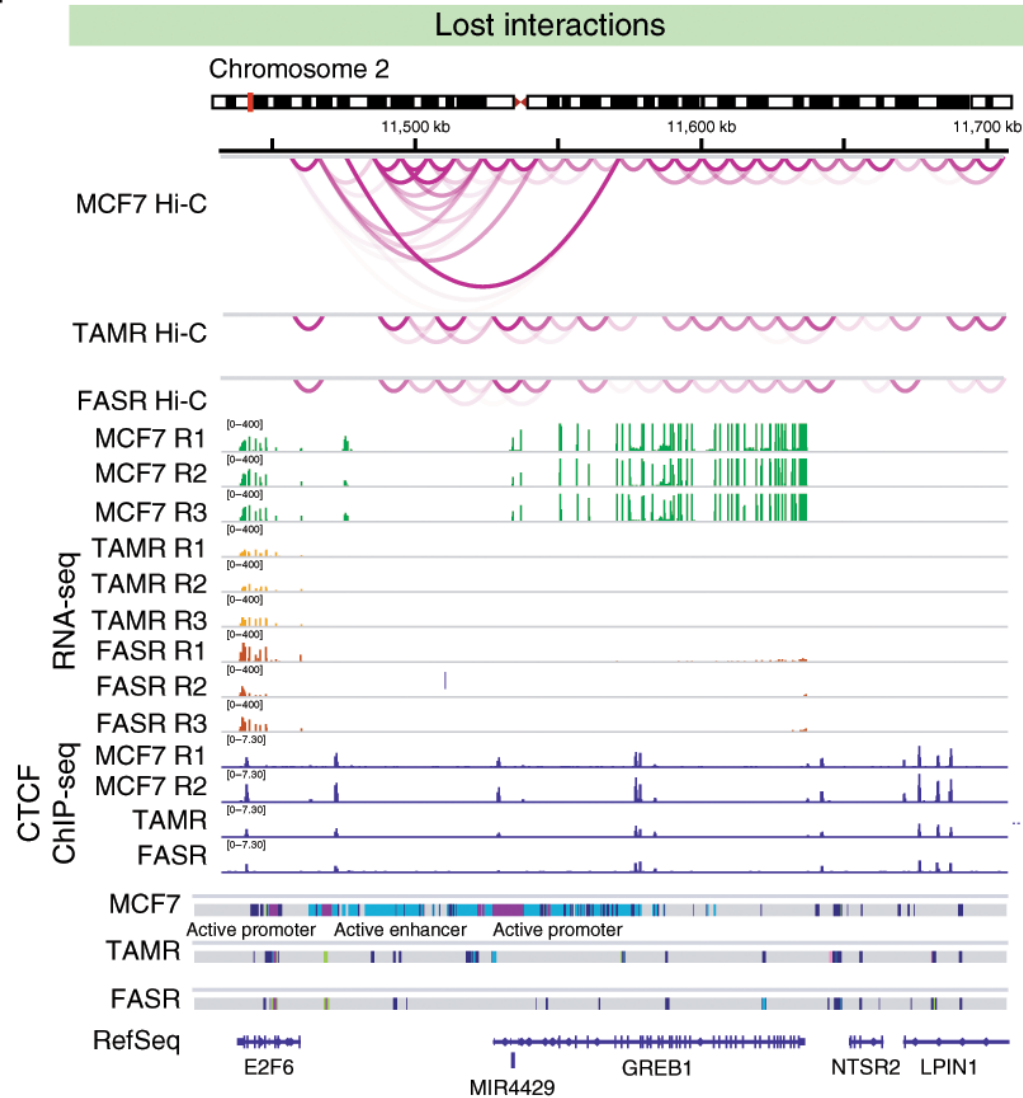
b | Mice lacking the ZRS *Shh* limb bud enhancer are born with severely truncated limbs.

Differential enhancer–promoter interactions and gene deregulation

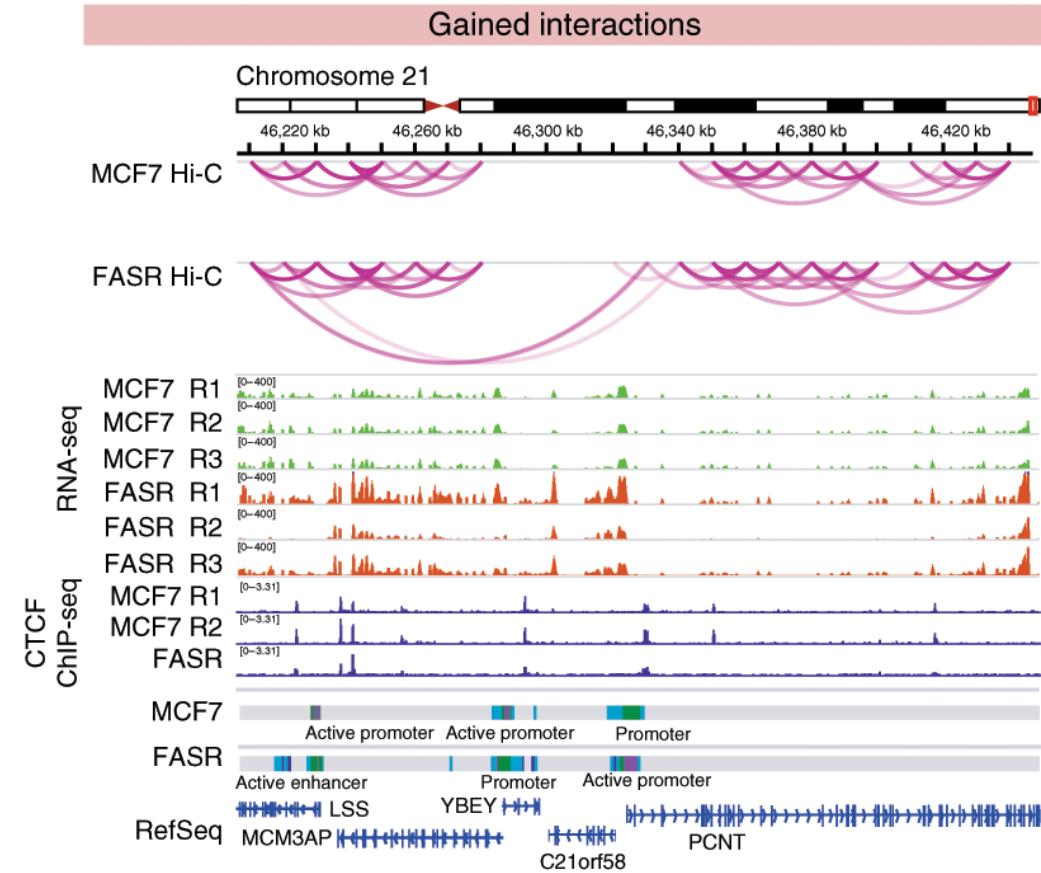


Differential enhancer–promoter interactions and gene deregulation

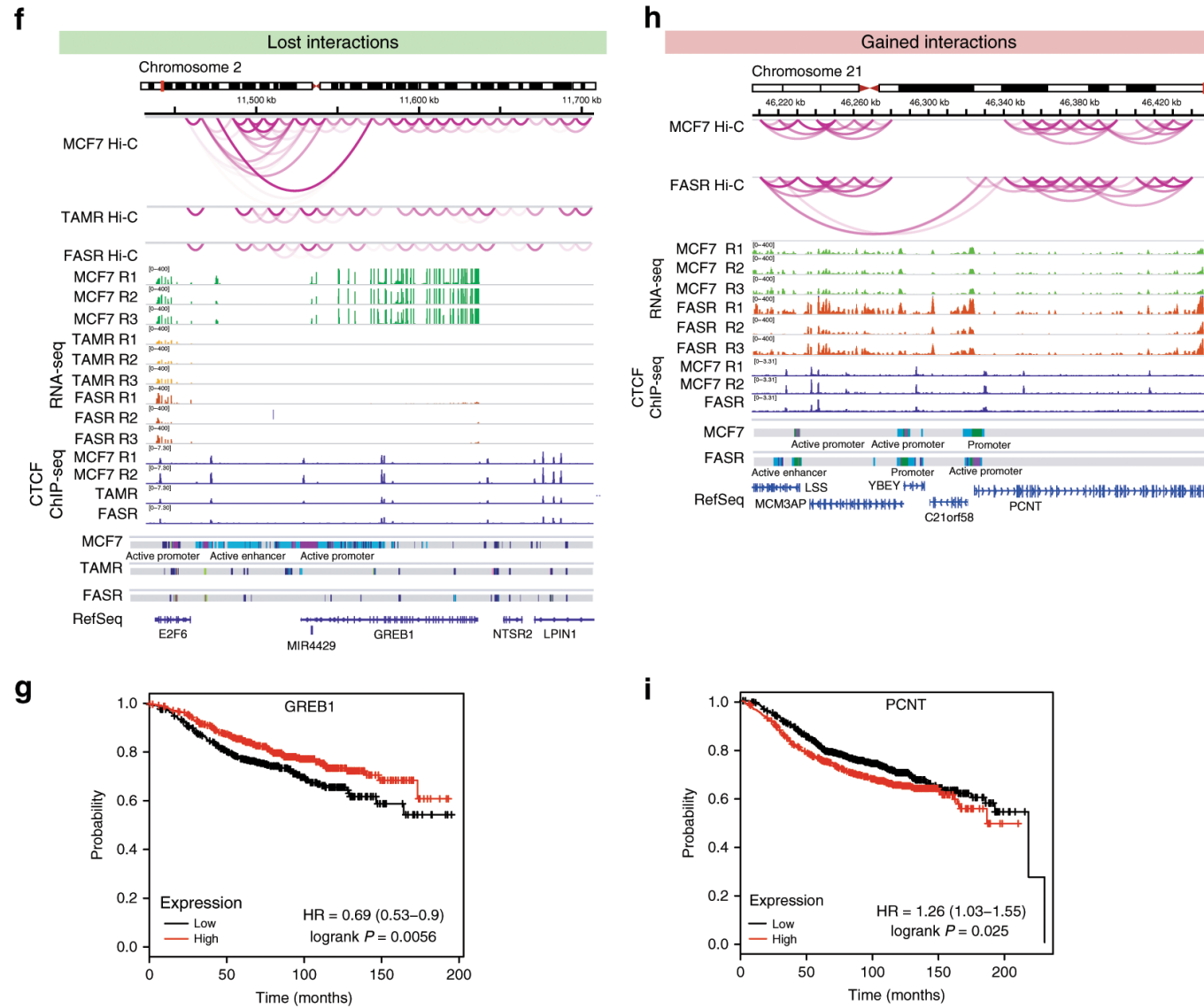
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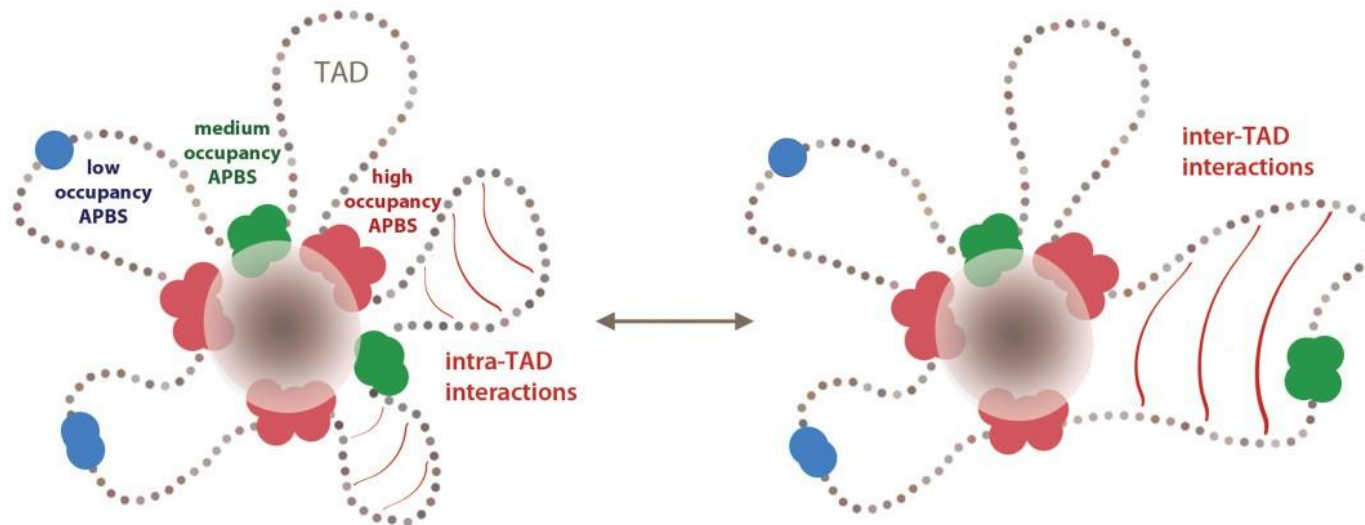
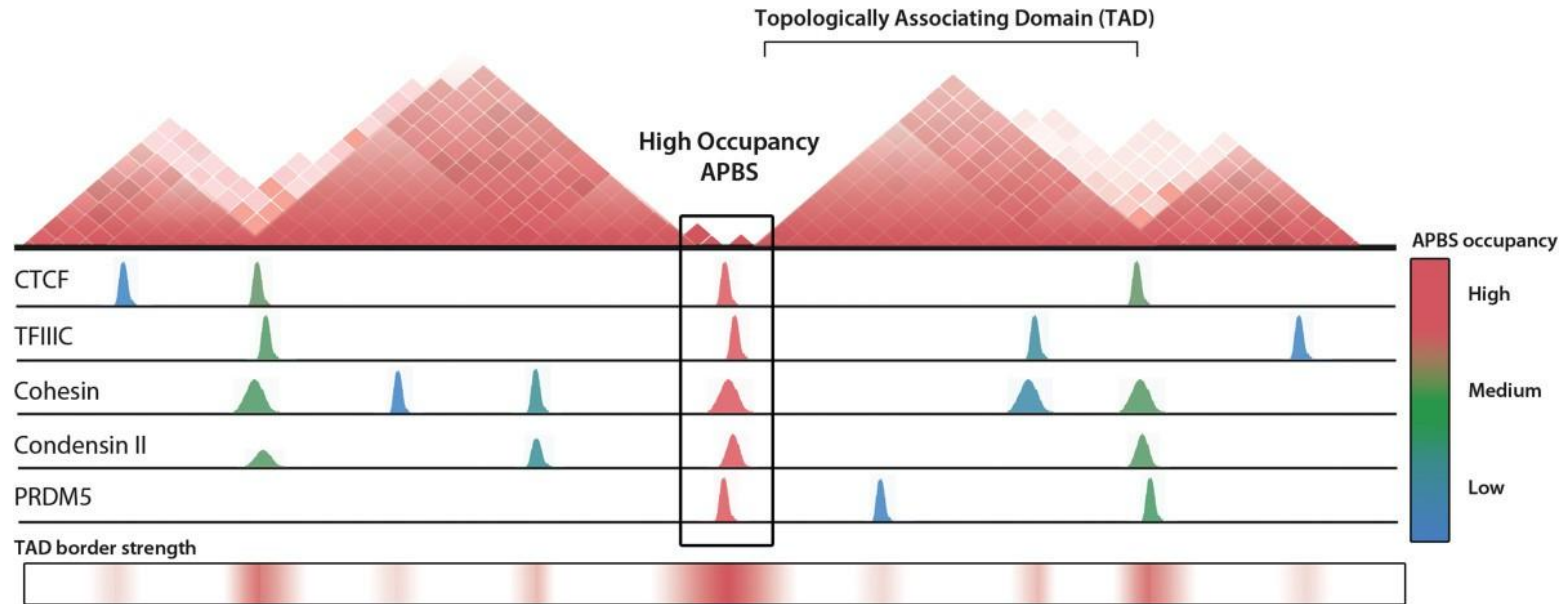
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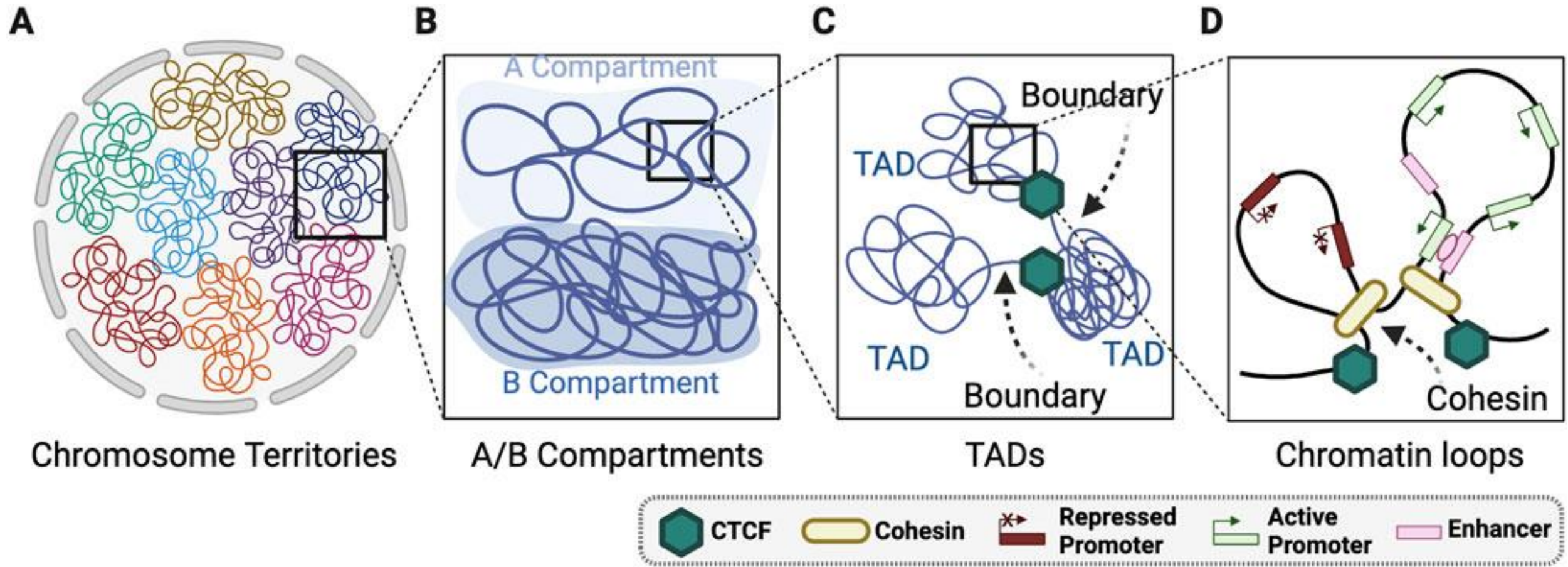
Differential enhancer–promoter interactions and gene deregulation



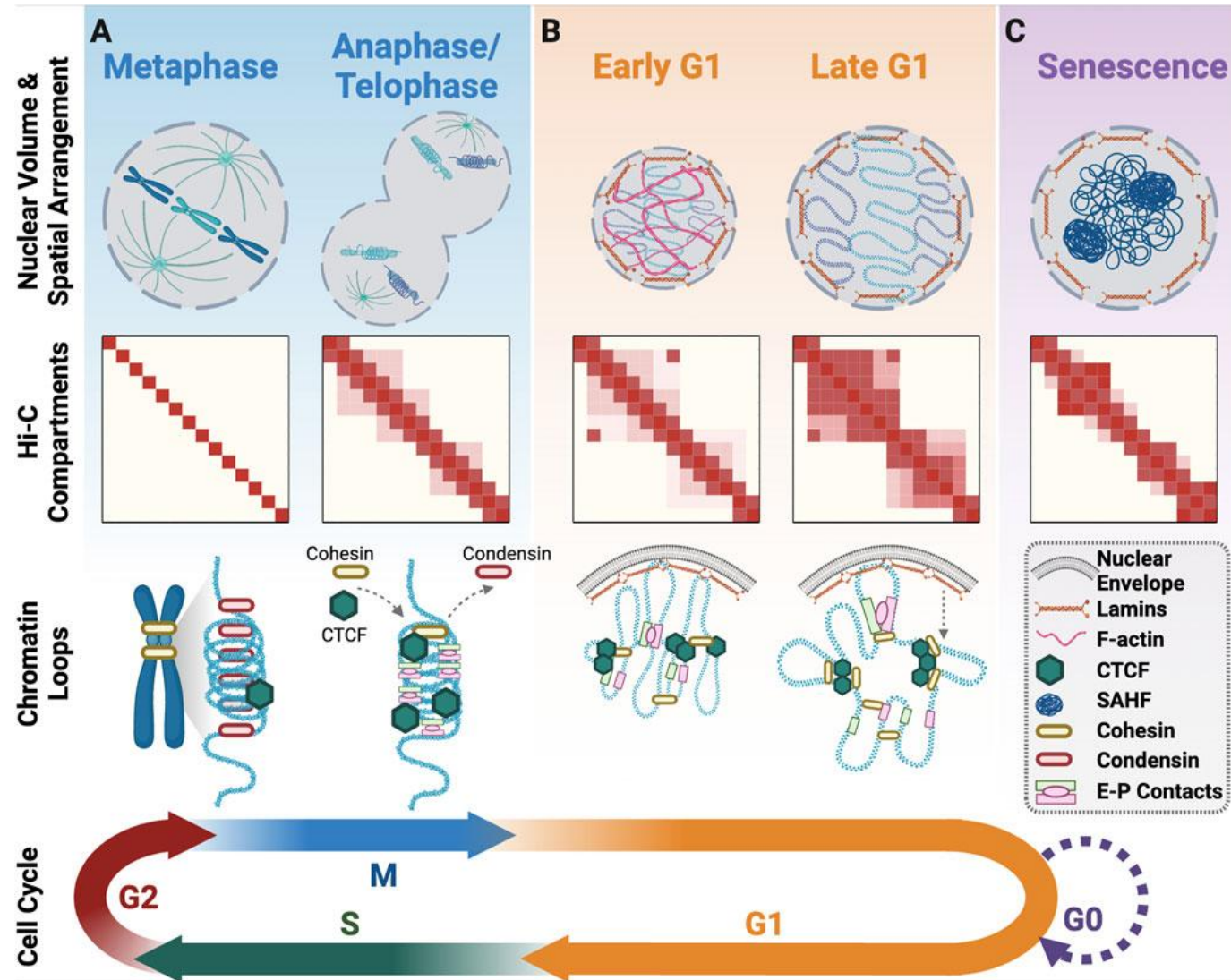
Combinatorial binding of architectural proteins shapes topological domain structure



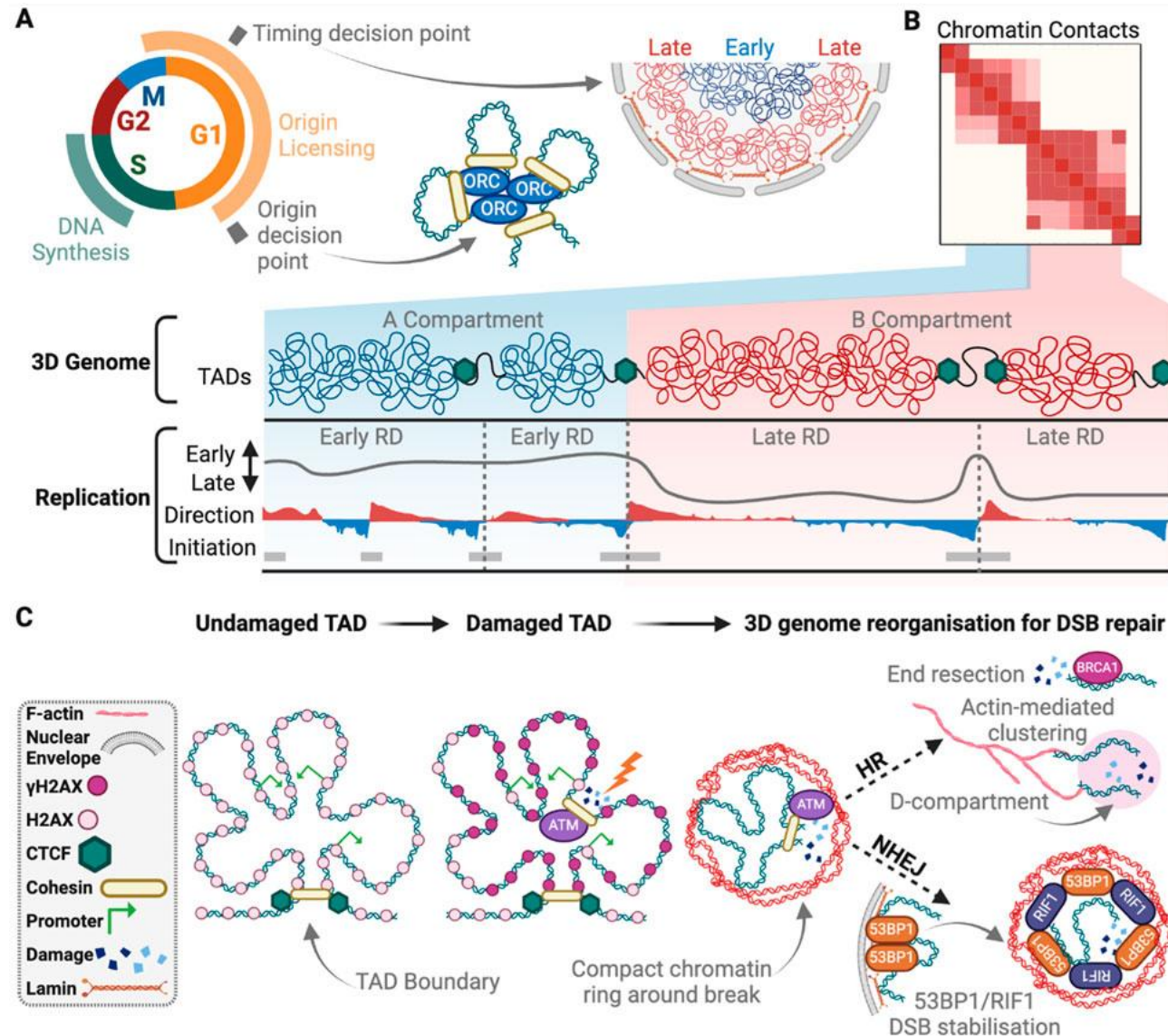
Layers of 3D genome organisation



3D genome organisation upon cell cycle exit and re-entry



The 3D genome in DNA replication and repair



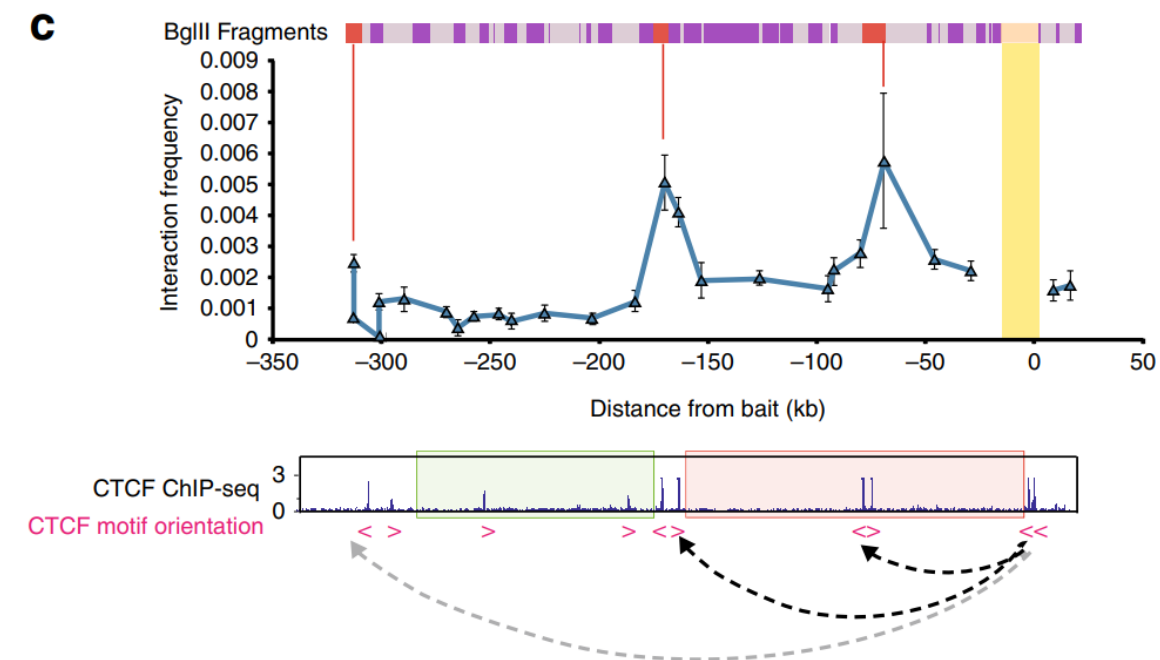
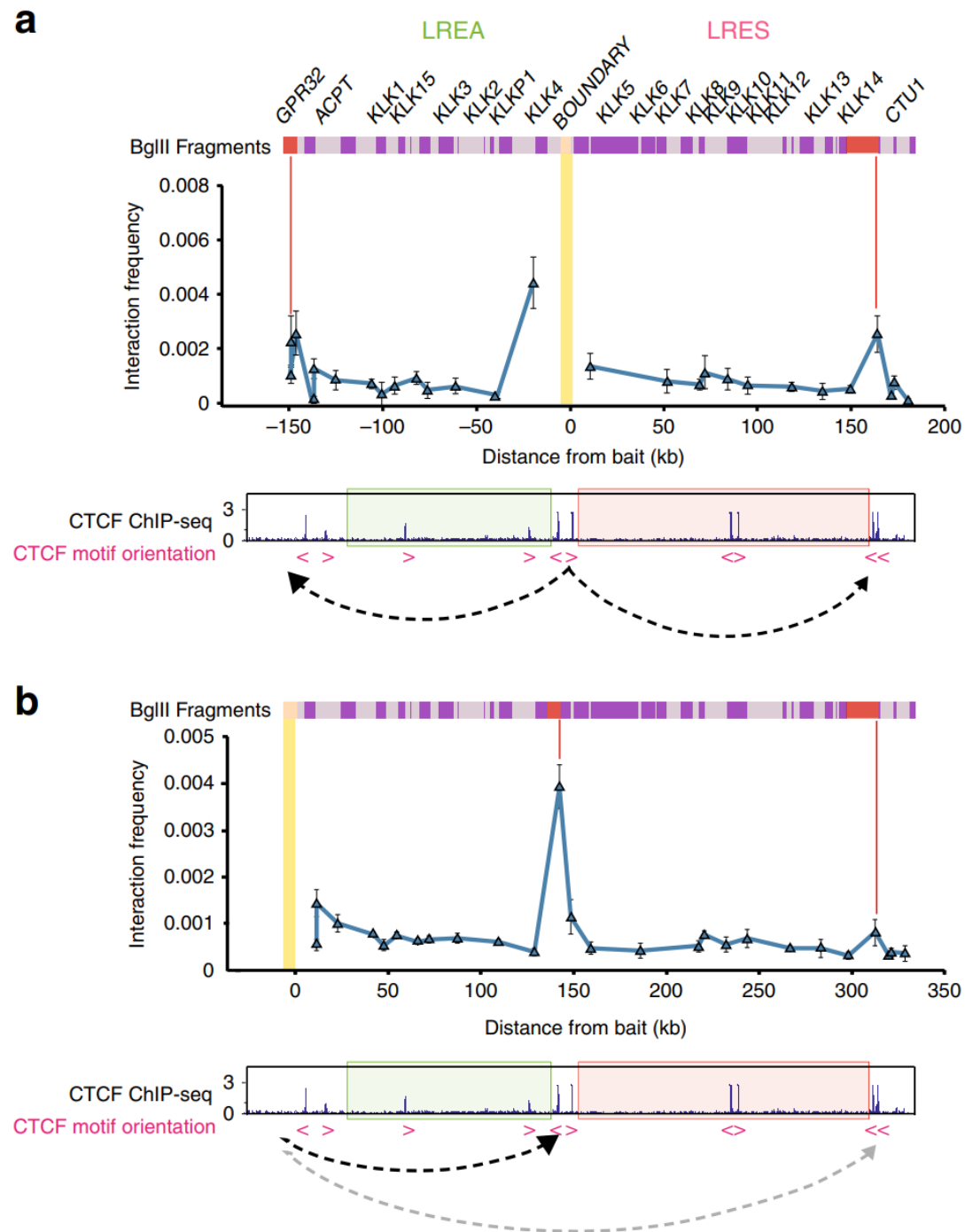


Fig. 2 3D structure of the *KLK* locus compartmentalises expression domains. **a** 3C-qPCR at the *KLK* locus. Bars above the graph illustrate BglII fragments across the region. The 3C bait fragment is indicated by the yellow bar and interacting fragments are shown by red lines (149.2 kb upstream from bait, 163.8 kb downstream from bait). The pink arrows indicate the direction of CTCF motifs. The dotted arrows demonstrate the loops indicated by the interaction data. Error bars represent standard error (SE). **b** As in **a**, utilising upstream interacting fragment from **a** as bait. Interacting fragments are 143.2 and 313.0 kb downstream from the bait. **c** As in **a**, utilising downstream interacting fragment from **a** as bait. Interacting fragments are 68.9, 169.8 and 313.0 kb upstream from the bait.

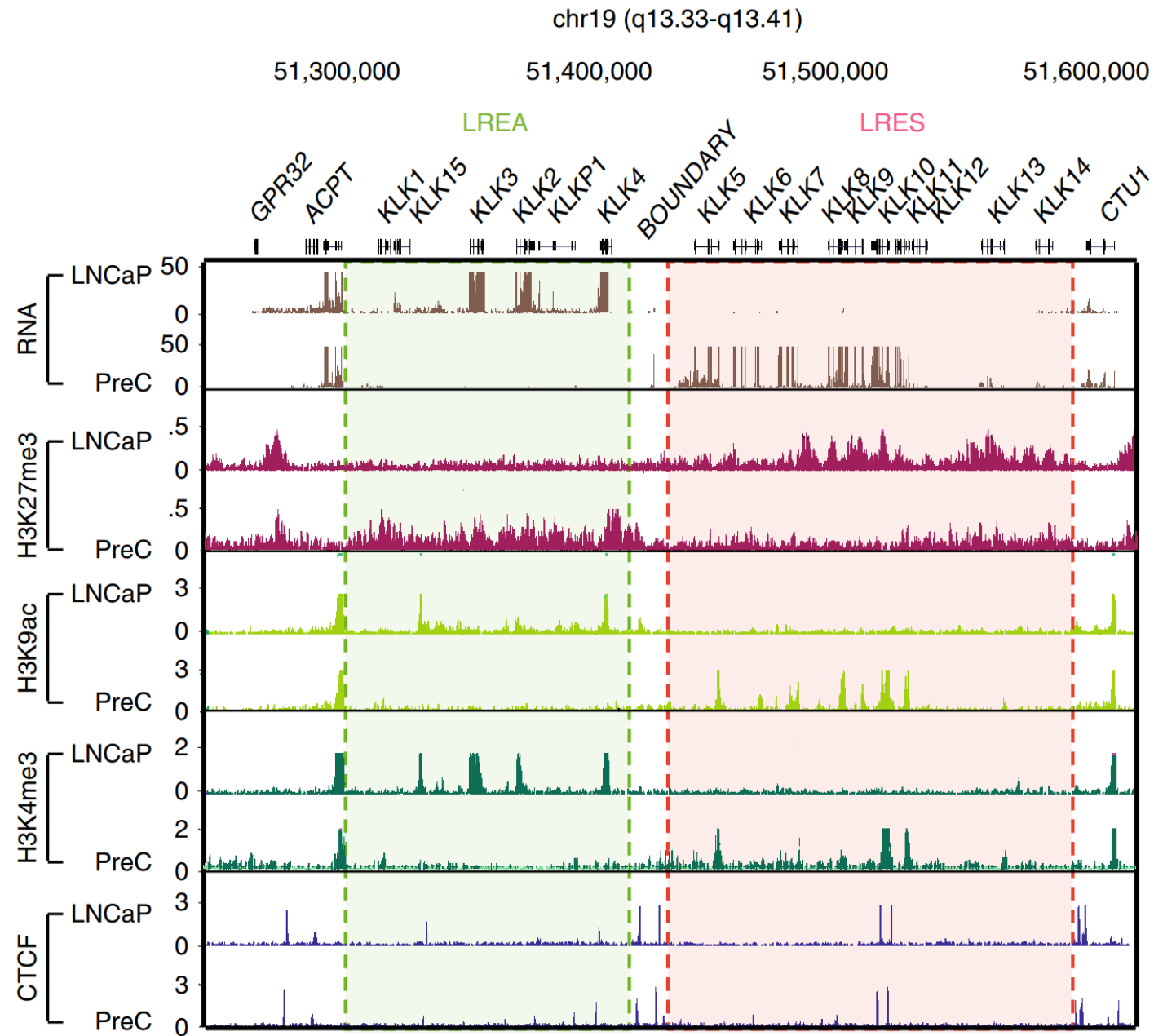


Fig. 1 Epigenome map of the Kallikrein locus in prostate cells. RNA-seq and ChIP-seq (H3K27me3, H3K9ac, H3K4me3 and CTCF) profiles of adjacent, transcriptionally active (green highlight) and silenced (red highlight) regions in LNCaP cancer cell line and normal PreC cells. CTCF ChIP-seq shows CTCF binding throughout the locus. Two CTCF sites are located at the boundary between the active and silenced regions.